

HACHINOHE AP, Japan

WMO: 475150

Lat: 40.552N

Lon: 141.467E

Elev: 162

StdP: 14.61

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	21.0	10.4	9.4	23.4	12.3	10.3	24.9	27.8	34.2	24.6	33.5	9.3	220	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.8	88.0	76.3	84.5	74.4	82.0	72.9	77.5	85.7	76.1	82.9	74.6	80.0	12.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.2	133.0	81.9	73.6	125.9	80.3	72.0	119.2	78.7	41.1	85.6	39.7	82.7	38.2	79.4	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.7	19.1	DB	15.1	93.3	2.3	2.7	13.5	95.3	12.1	96.9	10.9	98.5	9.2	100.5
			WB	14.0	79.6	2.1	1.6	12.5	80.7	11.3	81.7	10.1	82.5	8.6	83.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	30.5	31.7	37.6	48.0	57.2	62.7	70.5	73.6	67.7	56.5	45.0	35.0	(6)
(7)		DBStd	15.96	4.56	5.29	5.99	6.61	6.49	5.69	6.49	5.66	5.65	5.51	6.53	5.26	(7)
(8)		HDD50	2282	605	512	386	115	13	0	0	0	0	9	175	467	(8)
(9)		HDD65	5576	1070	932	848	512	252	108	21	3	32	268	599	931	(9)
(10)		CDD50	2810	0	0	3	54	237	381	637	731	530	210	26	1	(10)
(11)		CDD65	629	0	0	0	1	11	39	193	269	112	4	0	0	(11)
(12)		CDH74	3724	0	0	0	13	84	181	1163	1734	541	8	0	0	(12)
(13)		CDH80	1016	0	0	0	2	12	33	331	517	121	0	0	0	(13)
(14)	Wind	WSAvg	9.1	9.9	10.1	10.7	10.8	9.4	7.8	7.5	7.7	8.0	8.7	9.4	9.7	(14)
(15)	Precipitation	PrecAvg	46.2	2.6	2.1	2.6	2.8	3.5	4.0	5.6	5.6	6.5	4.3	3.3	2.8	(15)
(16)		PrecMax	58.4	5.2	6.0	4.6	5.8	5.6	8.0	10.9	12.9	12.7	8.5	6.4	7.4	(16)
(17)		PrecMin	34.9	1.1	0.8	0.8	0.6	1.3	1.3	0.9	1.0	2.1	1.1	1.1	1.4	(17)
(18)		PrecStd	6.3	1.1	1.1	1.1	1.3	1.3	1.6	2.5	3.3	2.6	2.2	1.5	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	53.1	62.7	75.0	81.0	84.3	91.7	93.1	88.1	75.0	66.4	55.2	(19)
(20)		MCWB	41.0	45.3	49.8	57.4	62.2	68.8	77.3	78.7	76.6	64.6	57.5	50.1	50.1	(20)
(21)		2%	DB	42.7	46.7	56.9	67.9	75.5	78.9	87.7	89.4	83.4	71.5	62.3	50.1	(21)
(22)		MCWB	37.4	40.3	47.1	54.1	60.7	67.1	76.0	77.2	74.2	62.3	54.8	45.2	45.2	(22)
(23)		5%	DB	39.5	43.2	51.9	63.0	71.8	75.2	84.0	85.8	79.2	68.4	59.0	46.6	(23)
(24)		MCWB	35.0	37.2	44.1	50.8	59.0	65.2	74.2	75.7	70.8	60.7	52.6	42.1	42.1	(24)
(25)		10%	DB	37.5	40.7	48.2	59.2	68.1	71.8	80.5	82.4	76.8	66.2	55.7	43.2	(25)
(26)		MCWB	33.4	35.5	41.5	49.5	57.8	64.1	72.6	74.2	69.7	59.1	50.2	38.9	38.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.7	46.2	51.6	59.4	65.5	71.4	78.6	79.7	78.3	67.4	59.3	51.3	(27)
(28)		MCDWB	44.7	50.7	60.3	70.1	75.5	77.8	88.8	89.0	85.4	71.3	63.5	54.4	54.4	(28)
(29)		2%	WB	38.0	41.0	47.6	55.8	63.1	69.0	76.5	78.2	75.8	64.2	56.0	46.3	(29)
(30)		MCDWB	41.1	45.7	54.8	64.2	71.8	76.1	84.9	86.6	81.3	69.4	60.9	49.7	49.7	(30)
(31)		5%	WB	35.7	38.0	44.8	52.9	60.8	67.0	75.1	76.7	73.3	62.2	53.4	42.3	(31)
(32)		MCDWB	38.7	42.4	50.6	61.1	69.5	73.8	82.0	83.9	78.1	66.7	57.9	45.8	45.8	(32)
(33)		10%	WB	33.8	35.4	41.7	50.6	58.8	64.9	73.4	75.4	70.6	60.3	51.1	39.3	(33)
(34)		MCDWB	36.7	39.5	47.1	57.6	66.8	71.0	79.2	81.0	75.3	64.9	55.1	43.0	43.0	(34)
(35)	Mean Daily Temperature Range	MDBR	11.5	13.0	14.5	16.9	15.9	12.9	11.9	11.8	13.8	16.1	15.0	11.8	11.8	(35)
(36)		5% DB	MCDBR	13.9	17.5	20.7	22.7	22.3	18.6	17.3	17.0	16.7	18.1	19.4	16.2	(36)
(37)		MCWBR	10.6	12.8	14.1	13.0	11.3	9.3	8.0	7.7	8.8	11.1	13.7	12.9	12.9	(37)
(38)		5% WB	MCDBR	13.5	16.9	19.8	20.1	19.8	16.4	16.0	15.0	14.5	15.1	17.6	15.8	(38)
(39)		MCWBR	11.4	13.3	14.8	13.2	11.3	9.2	8.2	7.3	8.3	10.5	13.8	13.4	13.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.365	0.408	0.479	0.527	0.542	0.524	0.525	0.498	0.442	0.415	0.395	0.375	0.375	(40)
(41)		taud	2.062	1.940	1.855	1.828	1.869	1.997	2.066	2.153	2.265	2.214	2.161	2.105	2.105	(41)
(42)		Ebn at Noon	238	245	241	239	239	244	242	246	252	243	227	223	223	(42)
(43)		Edh at Noon	36	48	58	64	63	56	52	46	39	36	33	33	33	(43)
(44)	All-Sky Solar Radiation	RadAvg	505	729	1071	1410	1560	1530	1362	1307	1134	887	586	431	431	(44)
(45)		RadStd	23	60	78	127	163	115	153	160	91	77	43	30	30	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page